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Time Series Analysis and Markov Chain Process for explaining the main periodicities in Nisyros Volcano, South Aegean, Greece

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Geology Environment Sustainability

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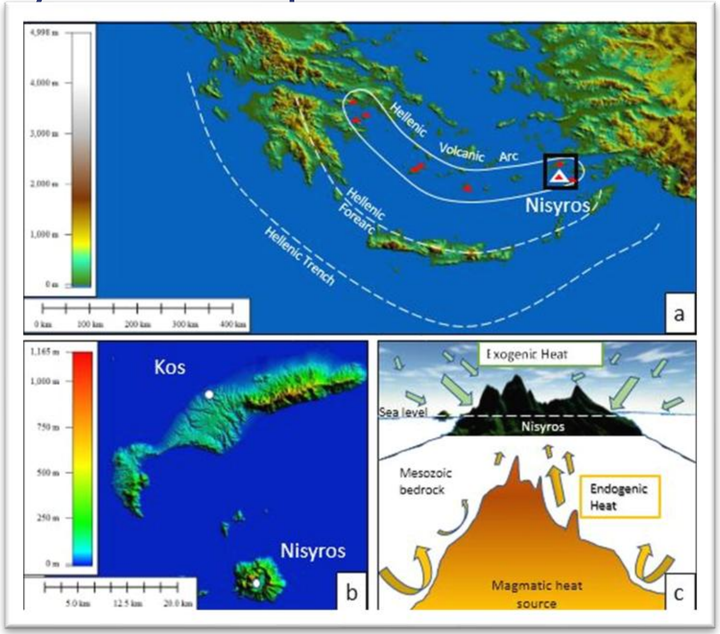
INTRODUCTION - METHODS

- Volcanic hazards monitoring is required in touristic places
- Volcanic surface temperature suffers from interferences
- Signal decomposition is required to identify different periodicities



METHODS

Time series decomposition & Markov CP

$$X(t) = L(t) + Se(t) + Sh(t)$$

X(t) represents the original time series of a variable,

L(t) is the long-term trend component,

Se(t) is the seasonal component, and Sh(t) is the short-term component.

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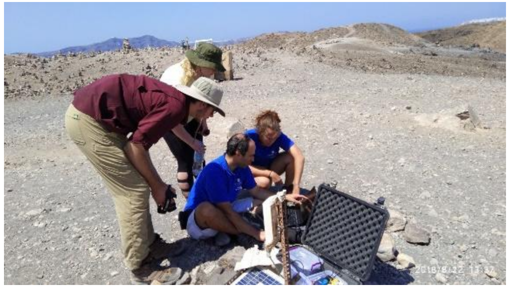
METHODS

MARKOV CHAIN (MC) PROCESS

- Breaks time series into sequence of segments
- Segments are identical temperatures
- MC identifies time distances between segments
- MC determines the periodicity of the distances



A volcanic monitoring station on a fumarole which is located at an active volcano (Thera) at the Hellenic Volcanic Arc in South Aegean Sea, Greece.

TIME SERIES DECOMPOSITION

- Synthesizes different time scale components into a meaningful linear regression
- Allows spectral analysis and determination of periodicity at each time scale component (Long, seasonal, short)

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METHODS - DATA

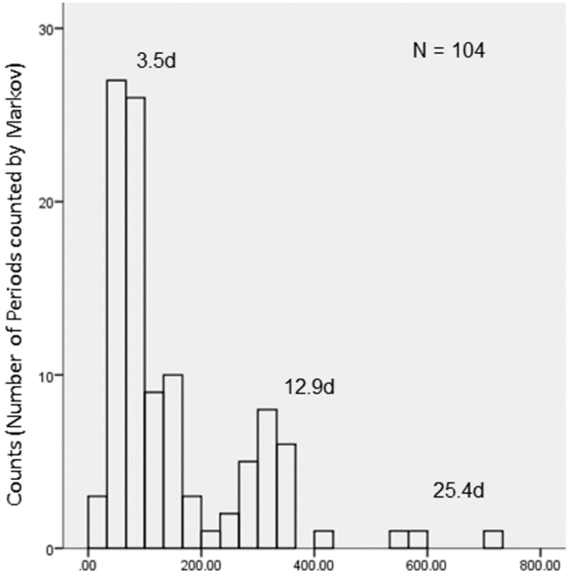
Volcanic temperature hourly data originated from:

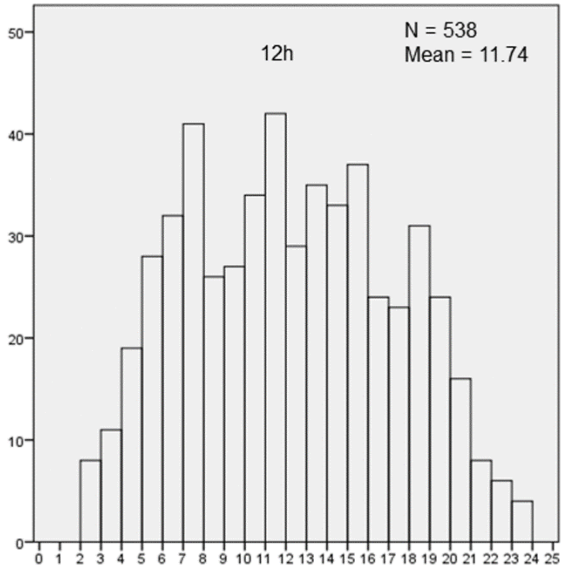
- **At Stephanos crater on Nisyros volcanic island**
- **Using an in-situ temperature dual probe Tinytag TGP-4520**

Thermal scanning and identification of best locations for continuous temperature monitoring

RESULTS

- **Long-term** component periodicities determined by MC process (left histogram)
- **Short-term** component periodicities determined by MC process (right histogram)





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CONCLUSIONS

- Multiple time scale periodicities contribute to noise related data
- Decomposition of temperature time series is prerequisite to avoid interferences of the data
- Short- and long-term data show cyclic phenomena that may be related to geological causes
- A cyclical operation of the existing micro-fracture system may permit a periodic heat release towards the volcanic surface
- This methodology may reveal other geophysical short-term oscillations originated by tectonic or other endogenic volcanic behavior.